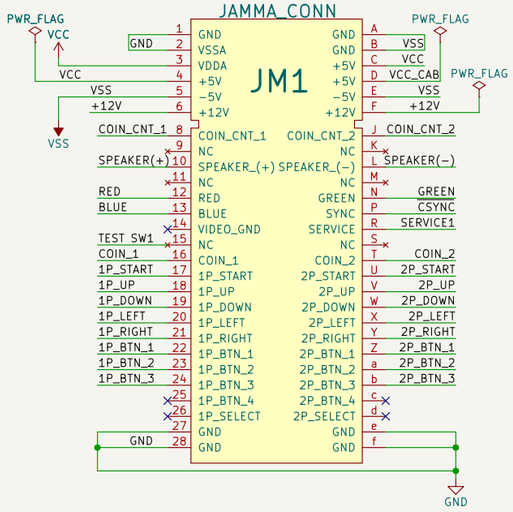
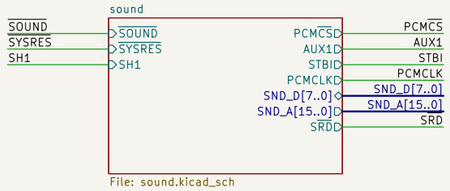
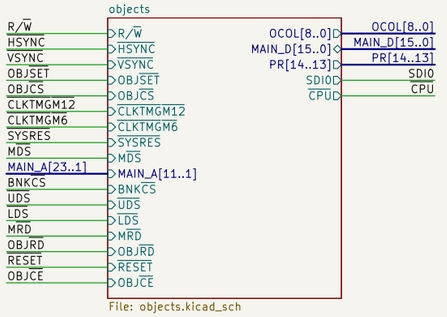
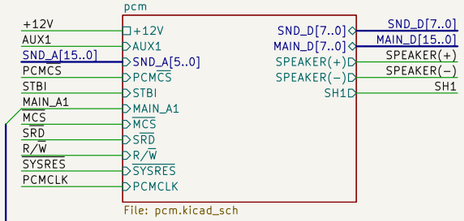
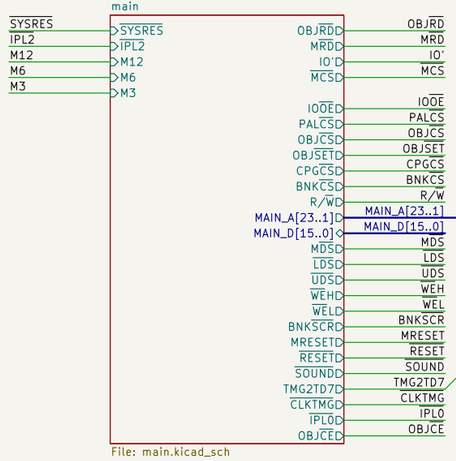
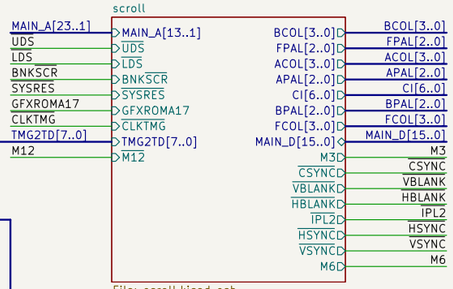


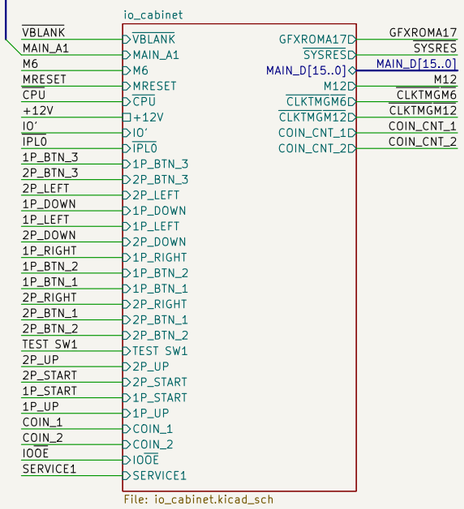



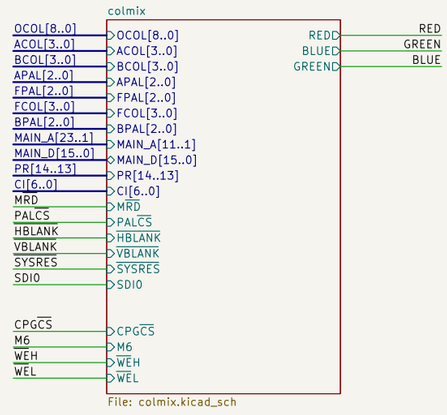









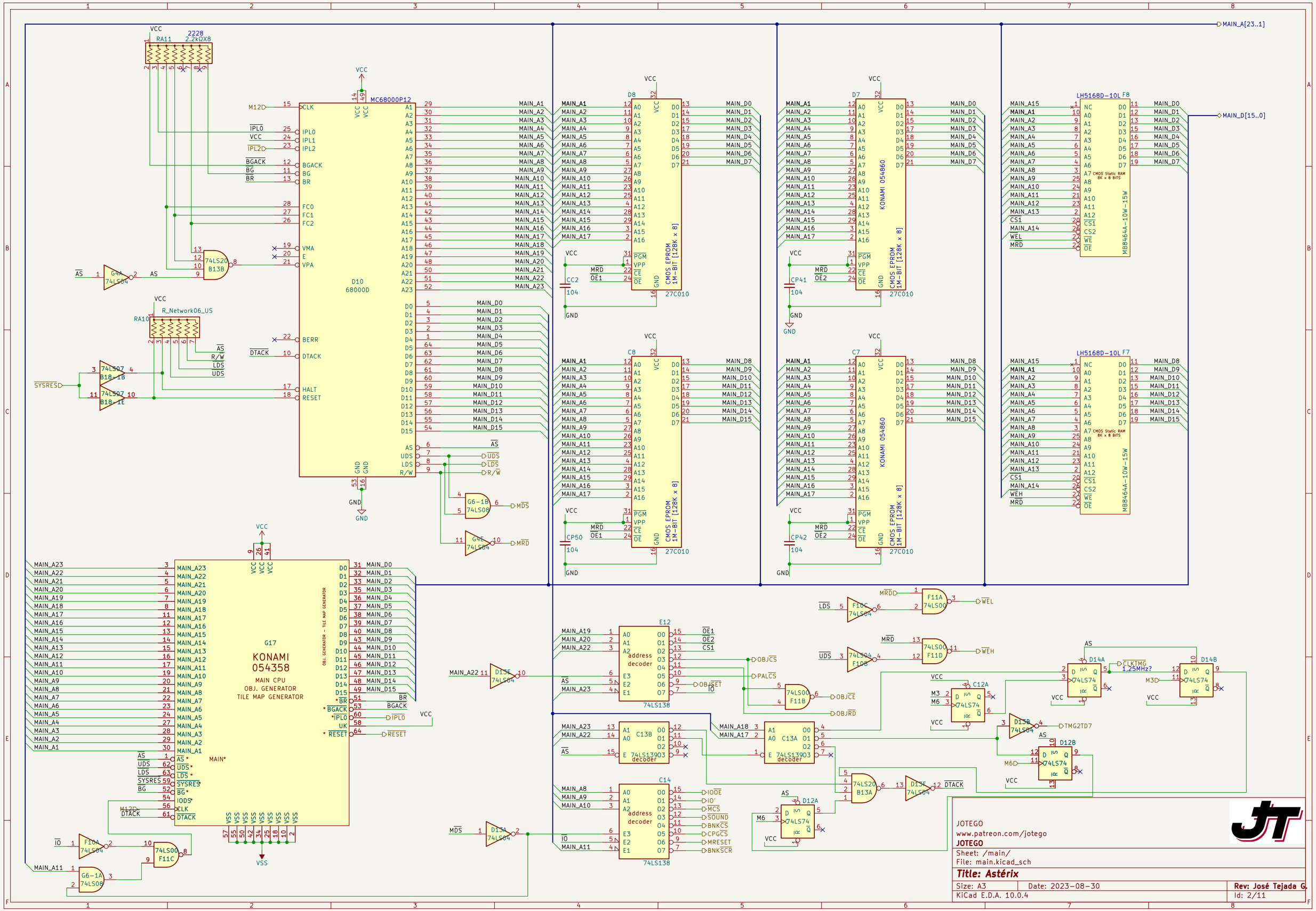




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**JOTEGO**

Sheet: /
File: asterix.kicad_sch

***Title: Astérix***

| Size: A3 | Date: 2023-08-30 | **Rev: José Tejada G.** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 1/11 |


MAIN_A[23..1]
MAIN_D[15..0]
RA11
2228
2.2kΩX8
MC68000P12
D10
68000D
M12D
IPL0
IPL1
IPL2
BGACK
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
DTACK
HALT
RESET
AS
R/W
LDS
UDS
SYSRES
R_Network06_US
RA10
74LS20
B13B
G4A
74LS04
74LS07
B18-1B
B18-1E
F10A
F11C
74LS00
G6-1A
74LS08
G6-1B
G4E
D8
D7
C8
C7
CMOS EPROM
1M-BIT [128K x 8]
27C010
KONAMI 054860
LH5168D-10L
F8
F7
MB8464A-10W-15W
CMOS Static RAM
8K x 8 BITS
CC2
CP50
CP41
CP42
104
G17
KONAMI
054358
MAIN CPU
OBJ. GENERATOR
TILE MAP GENERATOR
OBJ. GENERATOR – TILE MAP GENERATOR
MAIN*
E12
74LS138
address
decoder
C13B
C13A
74LS13903
C14
D13E
D13F
D13A
D13B
F10C
F10B
F11A
F11D
C12A
D14A
D14B
D12A
D12B
74LS74
CLKTMG
1,25MHz?
TMG2TD7
OBJCS
PALCS
OBJSET
IO
OBJCE
OBJRD
MRD
OE1
OE2
CS1
WEL
WEH
DTACK
MDS
MRDD
JOTEGO
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JOTEGO
Sheet: /main/
File: main.kicad_sch
Title: Astérix
Size: A3
Date: 2023-08-30
KiCad E.D.A. 10.0.4
Rev: José Tejada G.
Id: 2/11

MAIN_D[15..0]

VCC
CP2
104
VSS
SW42
TEST SW1
EVQ-QS5-07K

S2
COIN_2
COIN_1
SERVICE1
ICOIN_2
ICOIN_1
ITEST SW1
ISERVICE1
KONAMI_005273

S3
2P_UP
1P_UP
2P_START
1P_START
I2P_UP
I1P_UP
I2P_START
I1P_START
KONAMI_005273

S4
2P_LEFT
1P_LEFT
2P_DOWN
1P_DOWN
I2P_LEFT
I1P_LEFT
I2P_DOWN
I1P_DOWN
KONAMI_005273

S5
2P_BTN_1
1P_BTN_1
2P_RIGHT
1P_RIGHT
I2P_BTN_1
I1P_BTN_1
I2P_RIGHT
I1P_RIGHT
KONAMI_005273

S6
2P_BTN_3
1P_BTN_3
2P_BTN_2
1P_BTN_2
I2P_BTN_3
I1P_BTN_3
I2P_BTN_2
I1P_BTN_2
KONAMI_005273

B12
I1P_LEFT
I2P_LEFT
I1P_RIGHT
I2P_RIGHT
I1P_DOWN
I2P_DOWN
I1P_UP
I2P_UP
IOOE
MAIN_D0
MAIN_D1
MAIN_D3
MAIN_D2
74LS257

B14
I1P_BTN_1
I2P_BTN_1
I1P_BTN_2
I2P_BTN_2
I1P_START
I2P_START
I1P_BTN_3
I2P_BTN_3
IOOE
MAIN_D4
MAIN_D5
MAIN_D7
MAIN_D6
74LS257

C16
ICOIN_1
ICOIN_2
VCC
CPU
ISERVICE1
ITEST SW1
IOOE
MAIN_D8
MAIN_D9
MAIN_D11
MAIN_D10
74LS257

D16
VBLANK
IPLO
IOOE
MAIN_D12
MAIN_D13
MAIN_D15
MAIN_D14
74LS257

B17
ER5911
EEPROM
CS
CLK
DI
DO
RDY/BUSY
ORG
128X8 or 64X16
VSS

MAIN_D0
MAIN_D1
MAIN_D2
MAIN_D3
MAIN_D4
MAIN_D5
74LS174
hex flip flop
IO'
GFXROMA17

EMI222M
EMI1
+12V
A10
JP-A42
SJO
MRESET
COIN_CNT_2
COIN_CNT_1
Master/RESET generation for the whole system and watchdog timer
KONAMI
051550
CP1
104
Master /RESET generation for the whole system and watchdog timer coin counters handling.

C12B
SYSRES
MAIN_A1
D18A
74LS32

B11B
24.00000MHz
CC3
104
XTAL_OSC
M12
F12A
CLKTMGM12
F12B
M6D
CLKTMGM6

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Sheet: /io_cabinet/
File: io_cabinet.kicad_sch

***Title: Astérix***

| Size: A3 | Date: 2023-08-30 | **Rev: José Tejada G.** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 3/11 |

NOT POPULATED

CPU ADDRESS

CPU DATA TO READ GFX ROMs

GFX ROM ADDRESS

VRAM ADDRESS

VRAM DATA

ATTRIBUTE CODE FROM 0541567
LAYER A,B -H SCROLL?

G14
KONAMI
054156
TILE MAP GENERATOR

J14
KONAMI
054157
TILE DRAWING

LAYER A COLOR INDEX

FIX LAYER COLOR INDEX

LAYER B COLOR INDEX

GFX ROM DATA

COLOR OUTPUTS
053251

ATTRIBUTE CODE FROM 0541567
LAYER A, B - H SCROLL?

CPU DATA TO READ GFX ROMs

K12
955D05-06-07-08
KONAMI
MASK 4M - 16BIT
HN62414P
CMOS ROM

K16
955D05-06-07-08
KONAMI
MASK 4M - 16BIT
HN62414P
CMOS ROM

G11
MB8464A-10W-15W
CMOS Static RAM
8K x 8 BITS

G10
MB8464A-10W-15W
CMOS Static RAM
8K x 8 BITS

Conn_02x32_Row_Letter_First

G6-1C
74LS08

74LS07
B16A

74LS04
G4B

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Sheet: /scroll/
File: scroll.kicad_sch

***Title: Astérix***

| Size: A3 | Date: 2023-08-30 | **Rev: José Tejada G.** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 4/11 |

OBJROMA[18..0]

MAIN_A[11..1]

MAIN_A[11..1]

MAIN_D'[11..0]

OBJROMD[31..0]

MAIN_D[15..0]

RA[11..1]

RD[15..0]

AR[2..0]

BR[2..0]

HP[9..0]

PL[6..0]

NOT POPULATED

NOT POPULATED

J41
KONAMI
053245
OBJECT GENERATOR

OBJECT ROM ADDRESS BUS

OBJECT RAM ADDRESS BUS

OBJECT RAM DATA BUS

MAIN ADDRESS BUS

GENERATOR TO DRAWING

J42
053244

G8 SHARED BUFFER

G9

SRAM16K 45N

SRAM16K 45N

HM6116P-3
2kB RAM

HM6116P-3
2kB RAM

Conn_02x32_Row_Letter_First

74LS153

74LS153

74LS32

74LS04

74LS08

74LS00

74LS245

74LS245

74LS174

74LS174

CMOS ROM

CMOS ROM

CMOS ROM

CMOS ROM

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| | |
|---|---|
| Sheet: /objects/ | |
| File: objects.kicad_sch | |
| ***Title: Astérix*** | |
| Size: A3 | Date: 2023-08-30 |
| KiCad E.D.A. 10.0.4 | Id: 5/11 |

**Rev: José Tejada G.**

MAIN_D[15..0]
MAIN_A[11..1]
PR[14..13]
VCC
VSS
6M
M6
SDI0
CO[10..0]
CI[6..0]
OCOL[8..0]
CPCS
FCOL[3..0]
FPAL[2..0]
ACOL[3..0]
APAL[2..0]
BCOL[3..0]
BPAL[2..0]

PRIORITY
COLOR INPUTS
CHIP SELECT INPUT
COLOR OUTPUTS
J43
KONAMI
053251
PRIORITY CONTROL UNIT
30 pins top/bottom
20 pins left/right

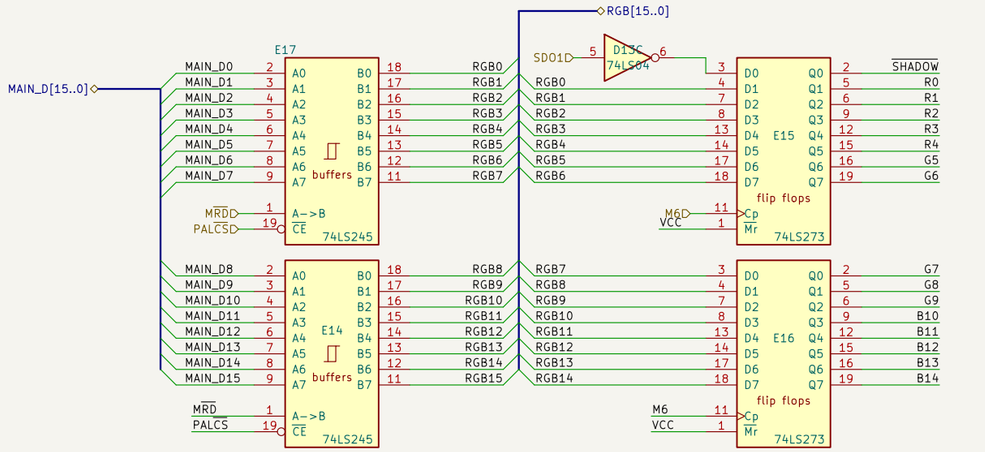
D15 Quad Mux 74LS157
E13 Quad Mux 74LS157
D17 Quad Mux 74LS157
SELECT_A-B
STROBE_G

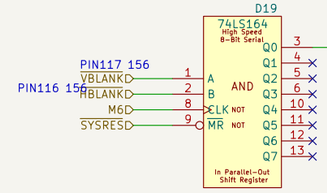
MCO[10..0]
RGB[15..0]
HM6116P-3
E18 MCM2018
E19 MCM2018
2kB RAM

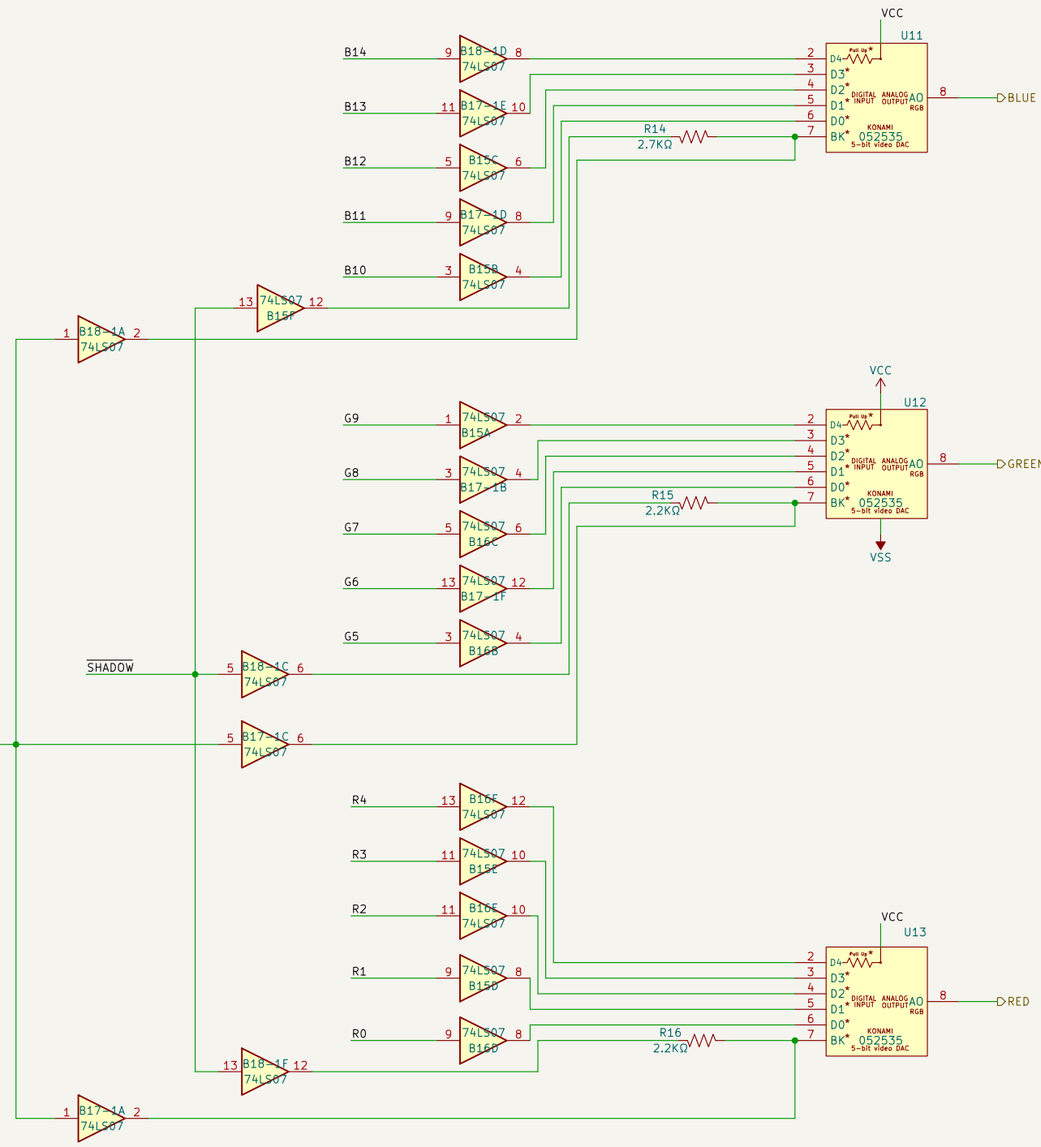
WELD
PALCSD
WEHD
D18D 74LS32
D18C 74LS32
MCOWRL
MCOWRH

rgb
RGB[15..0]
MAIN_D[15..0]
PALCS
MRD
SYSRES
VBLANK
HBLANK
M6
SDO1
REDD
GREEND
BLUED
RED
GREEN
BLUE
OK
File: rgb.kicad_sch

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**JOTEGO**
Sheet: /colmix/
File: colmix.kicad_sch
***Title: Astérix***
Size: A3 | Date: 2023-08-30 | **Rev: José Tejada G.**
KiCad E.D.A. 10.0.4 | Id: 6/11



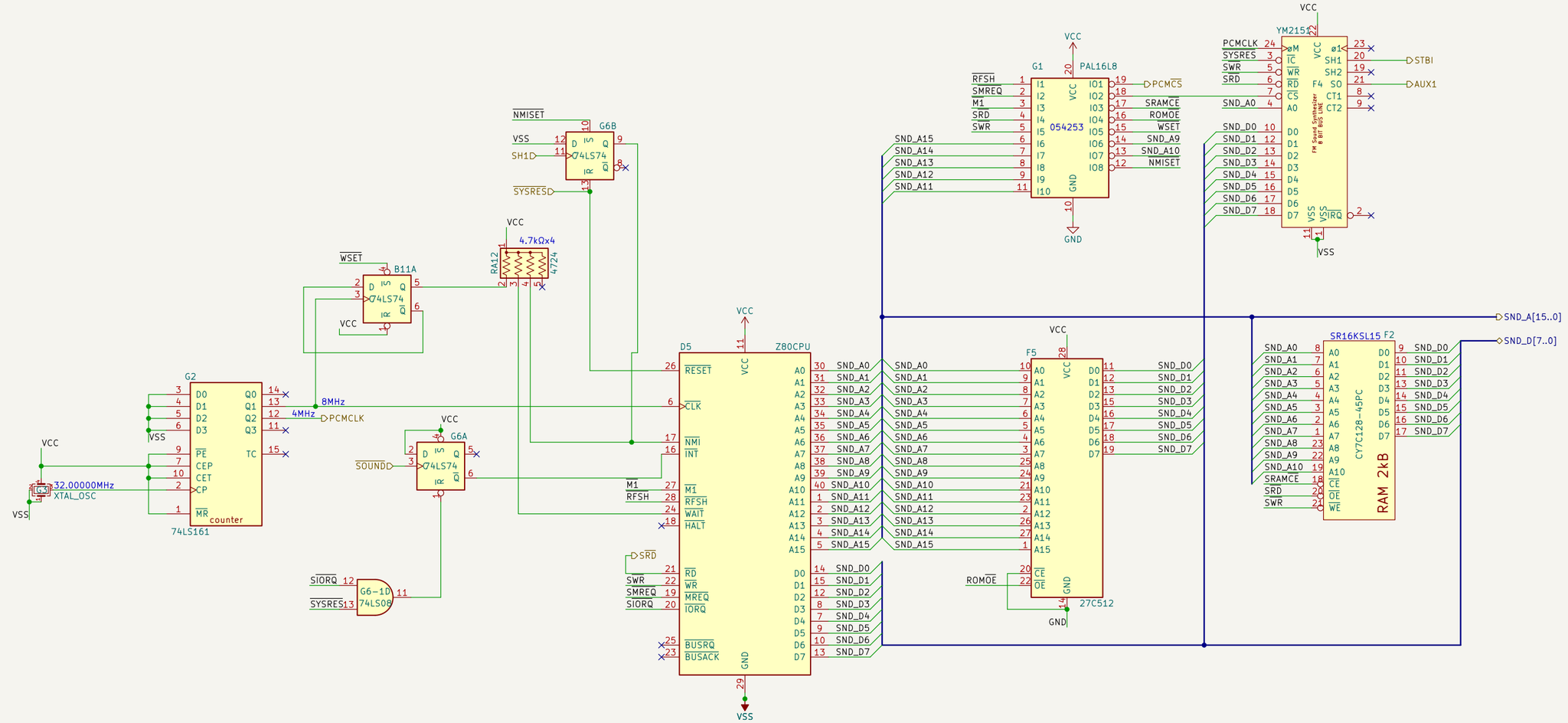




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**JOTEGO**

Sheet: /colmix/rgb/
File: rgb.kicad_sch

***Title: Astérix***

| Size: A3 | Date: 2023-08-30 | **Rev: José Tejada G.** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 7/11 |

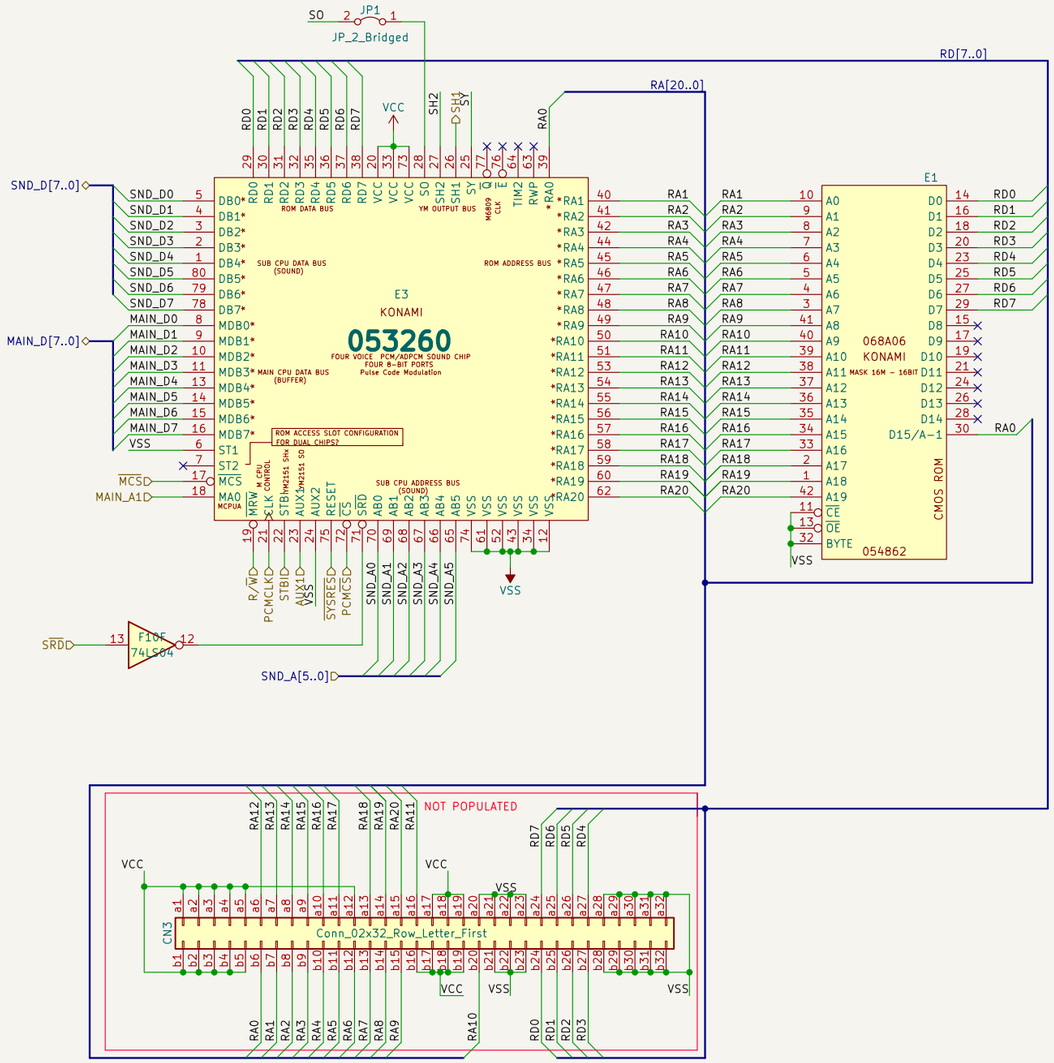


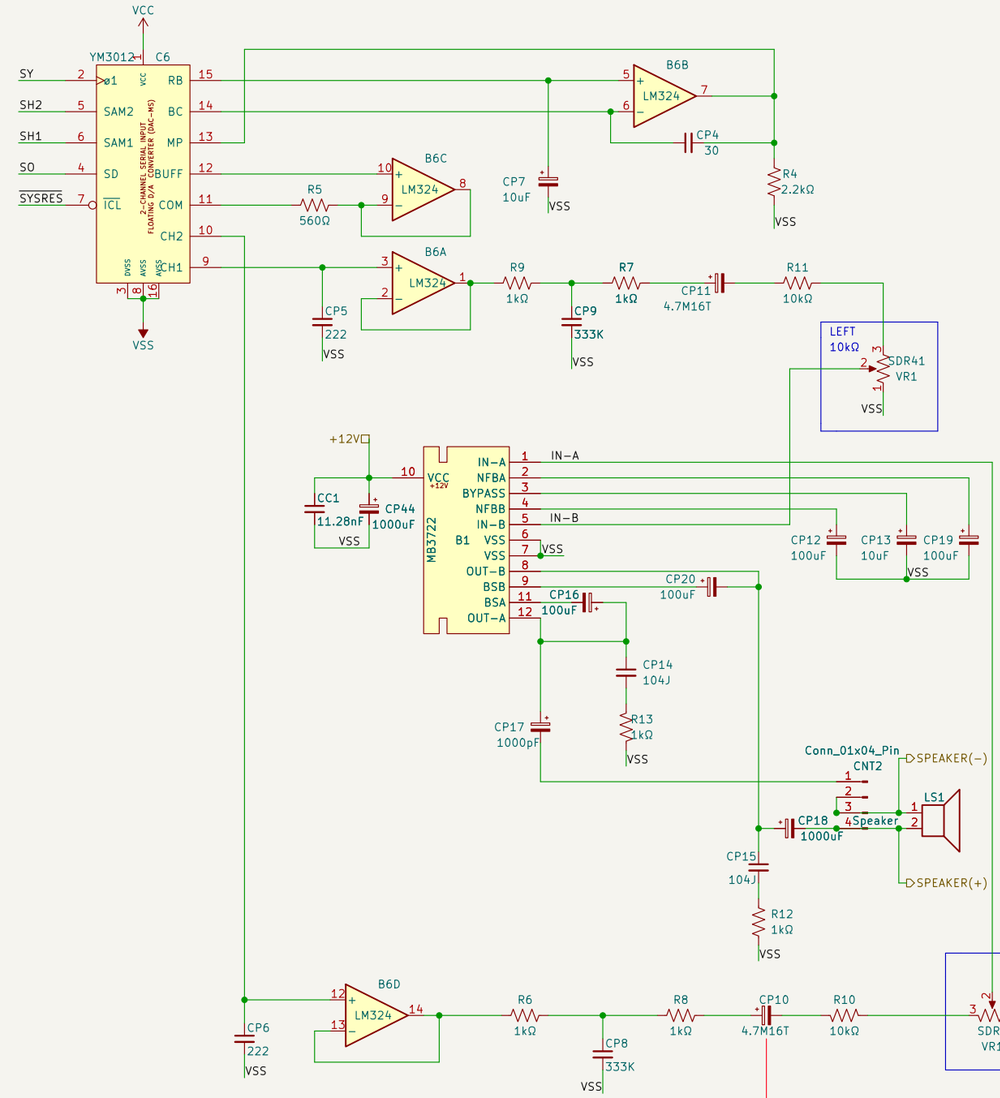

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**JOTEGO**

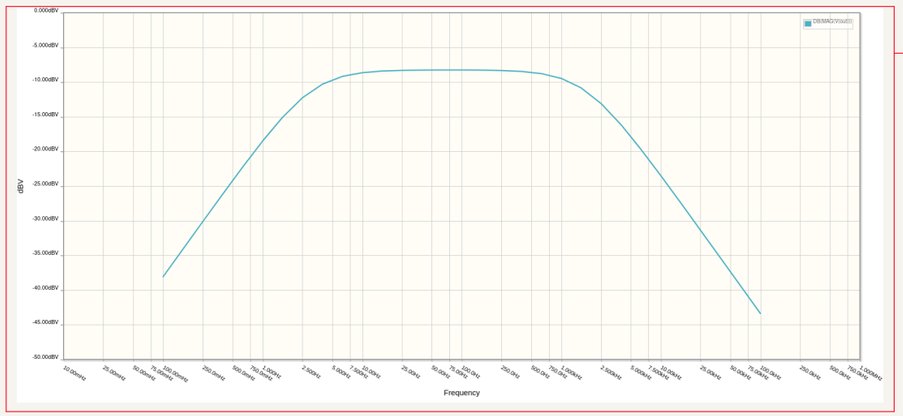
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***Title: Astérix***

| Size: A3 | Date: 2023-08-30 | **Rev: José Tejada G.** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 8/11 |




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**JOTEGO**

Sheet: /pcm/
File: pcm.kicad_sch

***Title: Astérix***

| | | |
|---|---|---|
| Size: A3 | Date: 2023-08-30 | **Rev: José Tejada G.** |
| KiCad E.D.A. 10.0.4 | | Id: 9/11 |

CP43 220uF
C41 104
C42 104
C43 104
C44 104
C45 104
C46 104
C47 104
C48 104
C49 104
C50 104
C51 104
C52 104
C53 104
C54 104
C55 104
C56 104
C57 104
C58 104
C59 104
C60 104
C61 104
C62 104
C63 104
C64 104
C65 104
C66 104
C67 104

VCC

G4G 74LS04
G5E 74LS00
G6C 74LS74
G6-1E 74LS08
G7E 74LS08
F10G 74LS04
F11E 74LS00
F12C 74LS74
4 V+
B6E
11 V-
LM324
B11C 74LS74
D18E 74LS32
D14C 74LS74
D13G 74LS04
D12C 74LS74
C12C 74LS74
C13C 74LS139
decoder
14 B18-1G 74LS07 GND 7
14 B17-1G 74LS07 GND 7
14 B16G 74LS07 GND 7
14 B15G 74LS07 GND 7
B13C 74LS20

VSS

G4G 74LS04
G4D 74LS04
G4F 74LS04
D15D 74LS04
G5B 74LS00
G5C 74LS00
G5D 74LS00
G7C 74LS08
G7D 74LS08

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**JOTEGO**

Sheet: /capacitors/
File: capacitors.kicad_sch

***Title: Astérix***

| Size: A3 | Date: 2023-08-30 | **Rev: José Tejada G.** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 10/11 |

# Asterix (arcade game)
# KONAMI XEXEX BASED HARDWARE

Asterix is a horizontal–scrolling beat'em up arcade game released in 1992 by Konami. It is based on the French comic series Asterix and Obelix. The player fights as either Asterix the Gaul or his best friend, Obelix, as they take on the 'might' of the oppressive Roman Empire. Asterix includes a variety of humorous fighting moves, which are demonstrated in the game's attract mode. The artwork and feel of Asterix remains true to its French source material, and the game's many humorous touches (such as the way defeated Legionnaires crawl away) will be instantly familiar to fans of the comic–book and cartoon series. Bonus levels – such as the chariot race that awaits at the end of the first level – introduce different gameplay elements to provide a break from the fighting. Despite this game being developed by a Japanese company, all text in the game is either in English or French.

**Konami Game Number : GX068**
**CPU : 68000**
**Other Chips : 054358**
**Objects Hardware : 053245 053244**
**Video Hardware : 054157 054156**
**Color Mixture : 053251**
**Sound Hardware : Z80**
**PCM : 053260**

*Source: https://en.wikipedia.org/wiki/X-Men_(1992_video_game)*
*http://www.system16.com/hardware.php?id=570&gid=830#830*

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**JOTEGO**

Sheet: /description/
File: description.kicad_sch

***Title: Astérix***

| Size: A3 | Date: 2023–08–30 | **Rev: José Tejada G.** |
|---|---|---|
| KiCad E.D.A. 10.0.4 | | Id: 11/11 |